Chapter 9: ZigZag

9.1. ZigZag Patterns

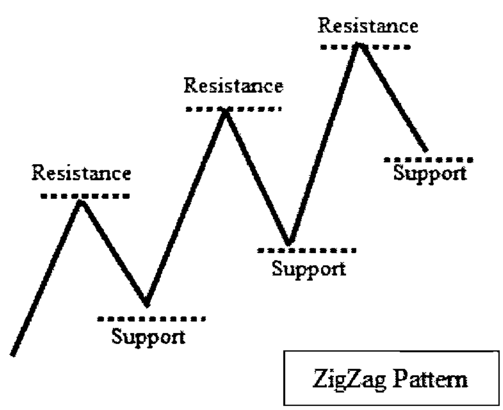
ZigZag Patterns

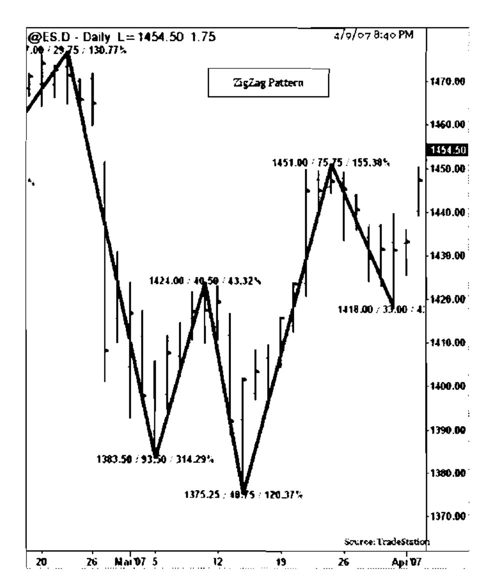
Market prices move in a non-linear format and form "peaks" and "troughs" to signal the end-of-trends and the begin-of-trends. Connecting these "peaks" and "troughs" generate a pattern called "ZigZag," which are used in Elliott Waves to detect various waves, using Fibonacci retracement to detect "price" clustering sequences. "ZigZag" patterns are also used to detect price patterns like "M" and "W" patterns. The construction of "ZigZag" patterns includes finding the key peaks and troughs with pivots and determining the "ZigZag" leg lengths, elapsed time and bar count. "ZigZag" patterns are eye pleasing and tempting but they provide no predictive analysis by themselves. They can be coupled with pattern recognition techniques or some other theories like overlaying on Renko charting to assist with trading.

"ZigZag" patterns do help filter out noise and compare each leg's size with previous swings to get trend strength. The "ZigZag" patterns use certain variables to validate the pattern such as minimum price movement for points or a percentage, and Average True Range for volatility measurement and "pivot" strengths.

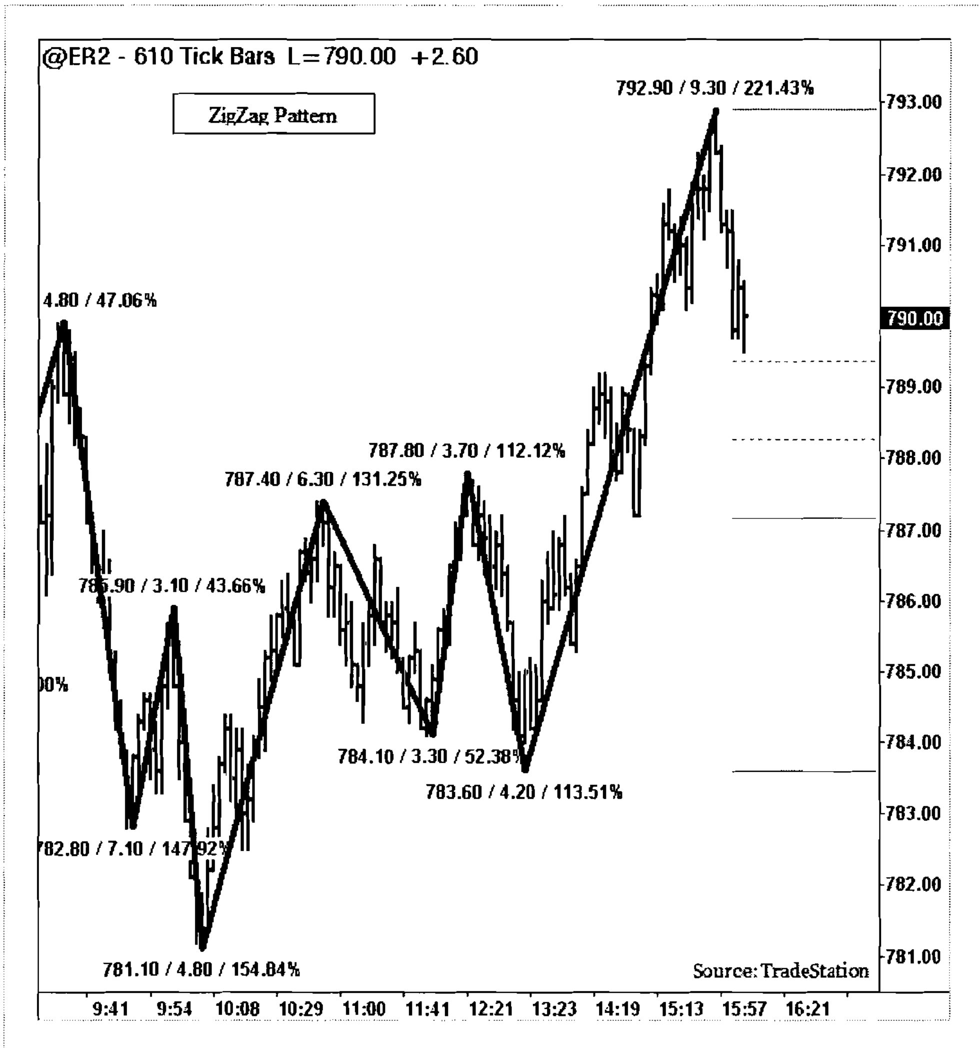
"ZigZag" patterns are used for price cluster generation to determine potential support and resistance areas. They are automatically drawn on charts to plot Fibonacci ratios for prior swings. J.D. Hamon and Michael Gur have published some theories about "ZigZag" and "Symmetric wave" theory trading.

Because "ZigZag" patterns are used only for trend indicators, there are no specific trading rules are provided.





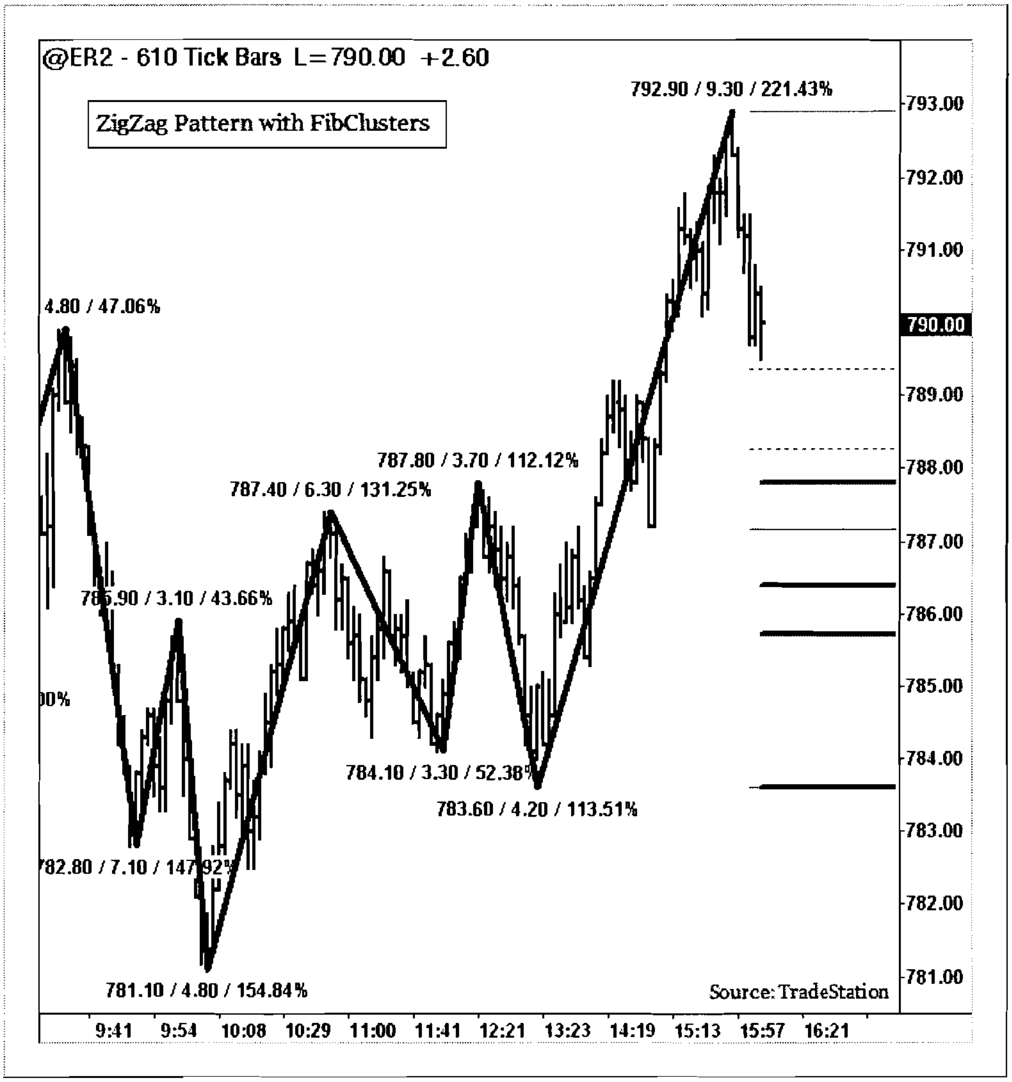
Trading ZigZag Pattern



Trading ZigZag Pattern

The example above illustrates a "ZigZag" pattern from the Russell Emini chart. This chart uses an Average True Range of 10 bars to determine the minimum "ZigZag" move required to compute the trend directions. At each "peak" or "trough," the number of points or price movement in each swing and the retracement percentage is plotted. For the last swing, the Fibonacci retracement levels are plotted to spot potential support and resistance areas.

Trading ZigZag with Fib. Clusters



Trading ZigZag FibClusters

"ZigZag" patterns do not have any predictability nature. They demonstrate the relative price movements and elapsed time to help other patterns or wave counts. However, there may be some price levels in each swing that could be important support and resistance levels. These price levels can be detected using Fibonacci price clustering methods, where each level is grouped with other price points within a price threshold. These clusters act as attractive support/resistance points for trading. In the chart above, price clusters are auto plotted using the ZigZag swings and a Fibonacci price clustering algorithm to point potential support and resistance areas (plotted horizontal lines on the right side of the chart).

9.2. Elliott Wave

Elliott Wave Pattern

In the late 1920's, Ralph Elliott discovered that markets have a rhythm and markets travel in a series of wave patterns. Elliott suggested markets move in 5 strength wave (impulse) and 3 correction waves.

The Elliott Wave theory is complex and is very subjective for each trader. This pattern occurs in all markets and in all time-frames. The Elliott Wave has few complex rules. The wave patterns are briefly described below:

Wave 1 (W1): The smallest of the Elliott waves and the start of the "fractal wave" pattern.

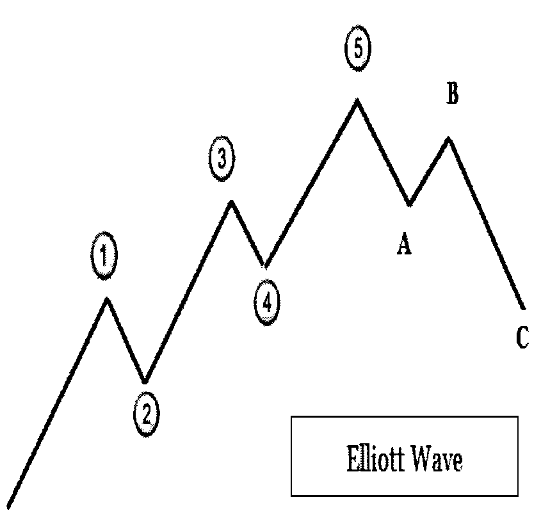
Wave 2 (W2): A retracement wave. Wave 2 should not trade below lows of Wave 1.

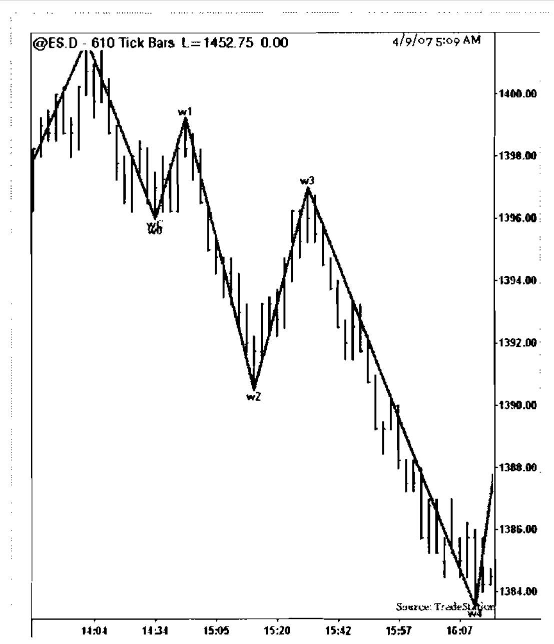
Wave 3 (W3): Longest and strongest of the entire wave count. Must trade above the "high" of Wave 1.

Wave 4 (W4): Usually called the "profit wave." Traders, who came in at Wave 1, will take profits. Wave 4 does not trade below Wave 2's high.

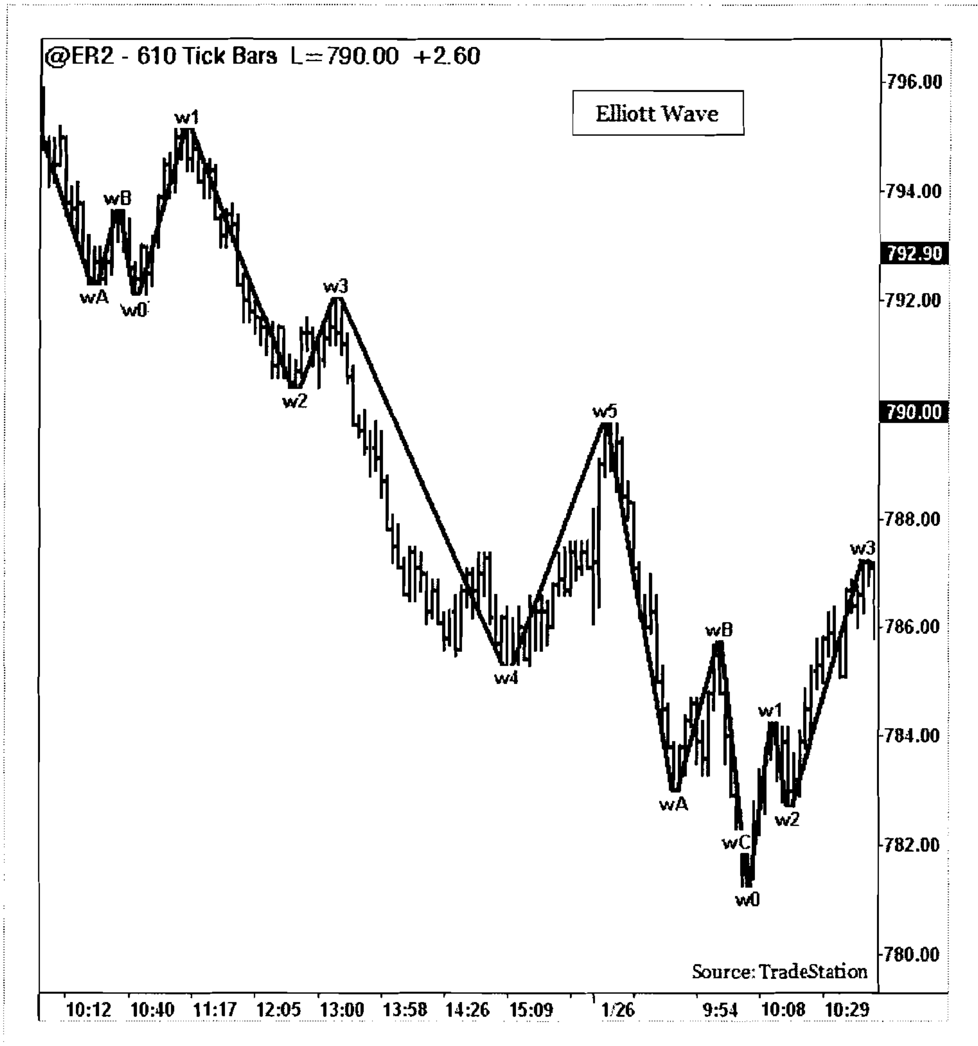
Wave 5 (W5): Also called "greed wave" or "overpriced wave." Exhaustion price movements occur before a serious correction.

After 5 wave patterns, markets form an ABC pattern in a three corrective waves fashion before another series begins.





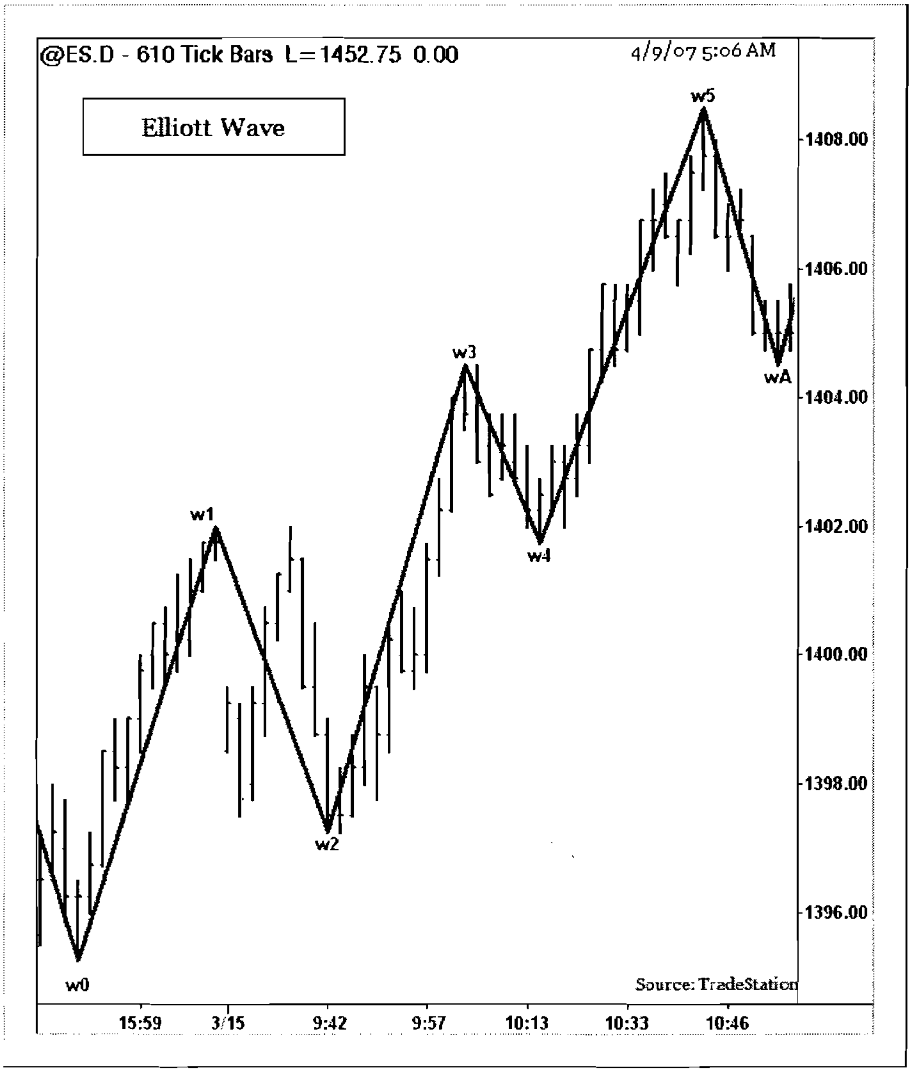
Trading Elliott Wave Pattern



Trading Elliott Wave Pattern

The basic Elliott Wave theory dictates that markets move in waves in the direction of primary trends. There will be 5-Action and 3-Reaction waves to complete a cycle. There are many variations and key aspects of these waves. However, the main concept is 5 "waves" up/down followed by 3 down/up cycles. Wave ranges can be determined by Fibonacci numbers for time/price ranges. Some traders plot "wave" channels to determine the "entry" and "target" areas. The Elliott Wave theory is very subjective and requires a very careful understanding of the entire wave theory thesis to be successful.

Trading Elliott Wave Pattern



Trading Elliott Wave Pattern

The Elliott Wave suggests that markets are always in progress and they correspond to 5 "up waves" and 3 "down waves" in a trend direction of "up or down." The above chart shows major waves consisting of smaller, minor waves in each. Trading Elliott Wave is not very easy. Nevertheless, when the rules are noted and the key origin wave (W0-W1) is found, then Elliott trading could be carefully applied. The example above illustrates an Elliott Wave from S&P 500 futures chart. Once W1 is identified, look for a retracement to W2. Most Elliotticians trade W3 and W5 as their identification becomes easier to follow and establish fixed rules. W3 is the longest wave. The W4 retracement stops above W1. W5 will be an extension wave of W3. There are many other Elliott Wave trading rules that can be found in other trading books and websites.

9.3. Crown Pattern

Crown Pattern

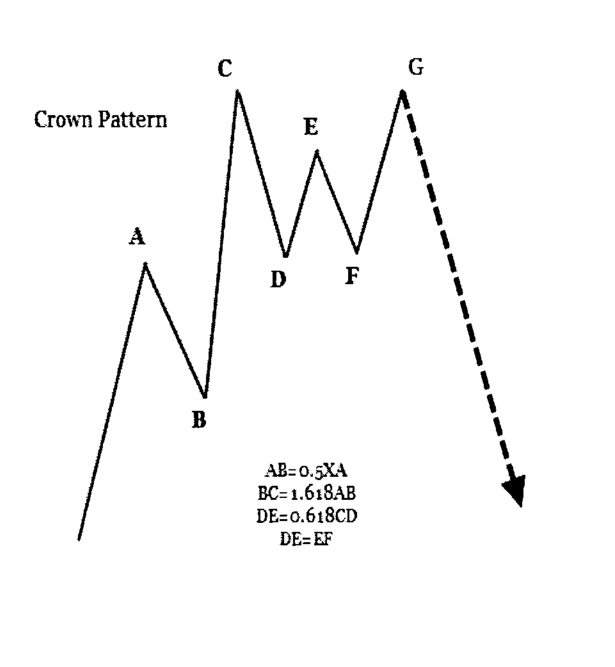
"Crown" patterns are a variation of skewed "Head and Shoulders" patterns, and they incorporate Fibonacci ratios in its formation. "Crown" patterns form at the end of market rallies or market declines suggesting a major trend change. As the name suggests, "Crown" patterns look like a series of "spikes" formed with Fibonacci ratios compared to prior swings.

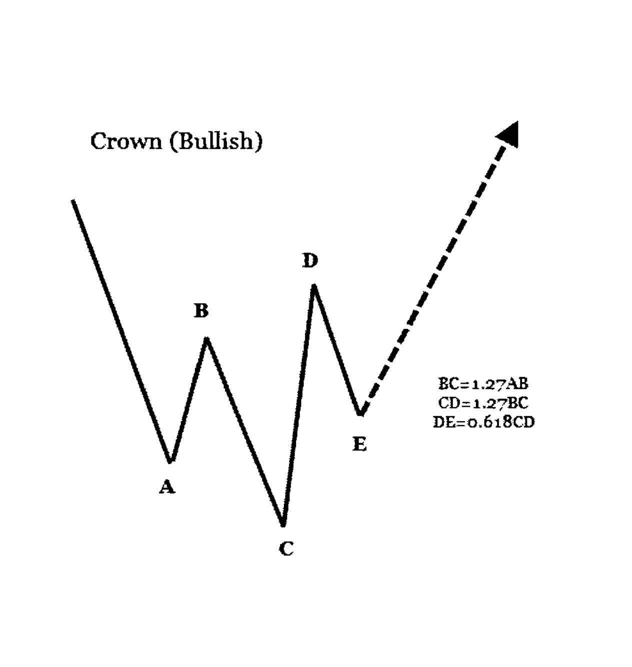
"Crown" patterns form in all markets and in all time-frames. They are moderately reliable, but need confirmation to trade. Occasionally, "Crown" patterns form multiple spikes like "ZigZag" in a short span of time before the pattern really emerges out of the channel.

Trade: "Crown" patterns have similar trading opportunities as "Head and Shoulders" patterns with few Fibonacci based rules. In bearish "Crown" patterns, trade setup is made below the low of the recent "swing low" and a "stop" placement is set above the recent "swing high." In bullish "Crown" patterns, trade setup is made above the high of the recent "swing high."

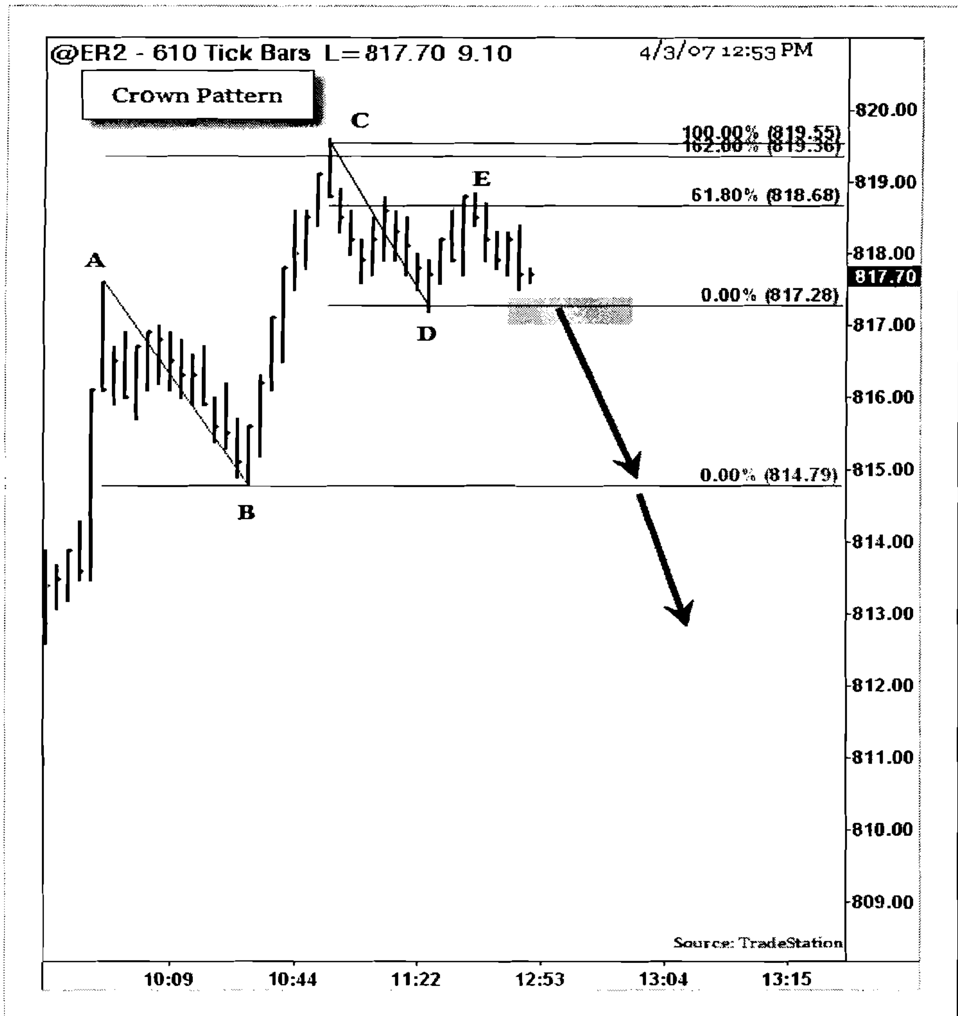
Stop: For short trades, if the market rally takes above the high of the pattern, the "Crown" will fail. Place an order 2 ticks above the "swing high" of the pattern. For long trades, reverse the previous criteria.

Target:
The "Crown" patterns are profitable and can be trailed with a 2-bar high for "short" trades and a 2-bar low for "long" trades. Set profit targets at major "swing lows" and major "swing highs."





Trading Crown (Bearish) Pattern

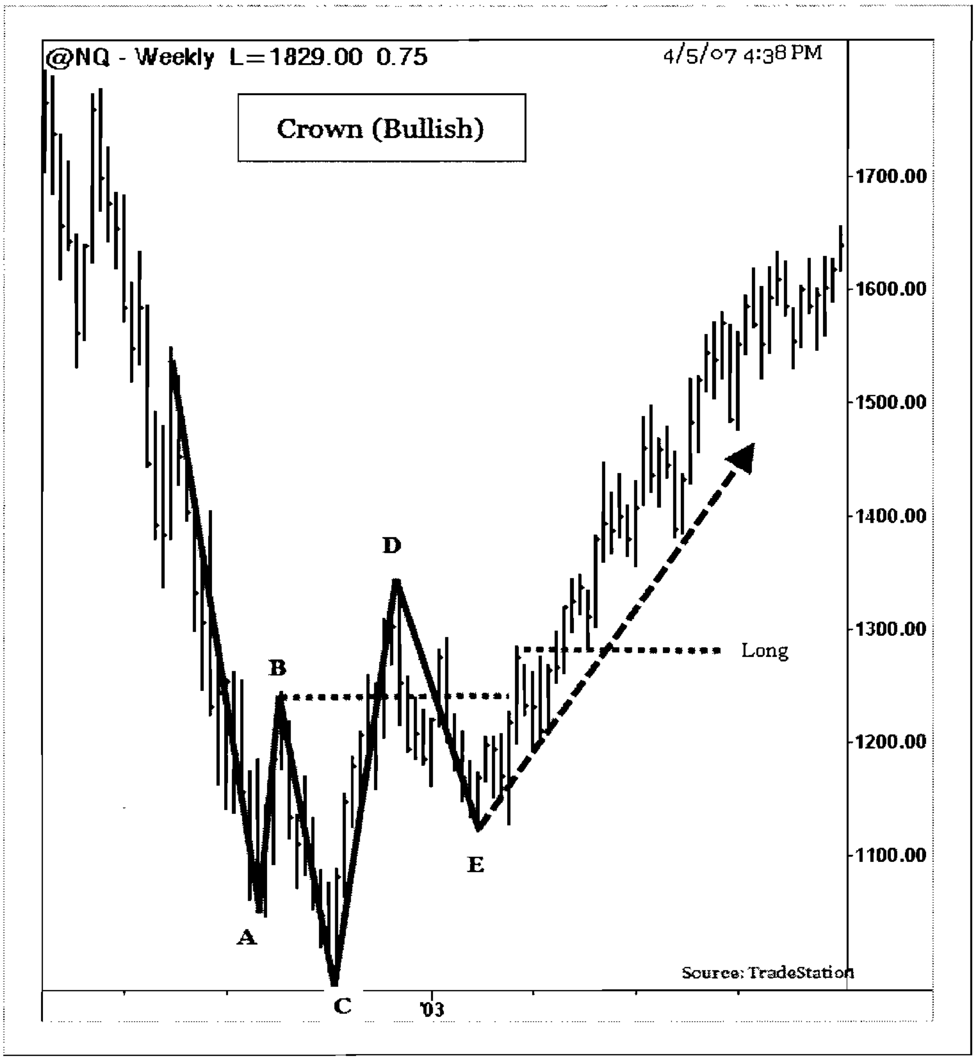


Trading Bearish Crown Pattern

The example above illustrates a "Crown" pattern from the Russell 2000 EMini futures chart. On April 04, 2007, during the morning session, ER2 traded higher and formed a "Crown" pattern based on Fibonacci ratios. The pattern continued its down trend after trading below recent "swing low" at 817. After a series of spikes, a "short" signal was triggered below the first "swing low" at "D" level.

1. Enter a "short" trade below the low at "D" level (817).
2. Place a "stop" order above the "E" level at 818.7
3. Target a major "swing low" prior to the "Crown" pattern formation at "B" level.

Trading Crown (Bullish) Pattern



Trading Crown (Bullish) Pattern

The example above illustrates a "Bullish" crown pattern from the Nasdaq futures (NQ) weekly chart. NQ made series of Fibonacci based swings to form a "Crown" pattern. The swing BC is 1.27 of AB and CD is 1.27 of BC and DE is 0.618 of BC. After series of spikes, look for a trend reversal to close above the first "swing high" at B.

1. Enter a "long" trade above level "B."
2. Place a "stop" order below the low of recent "swing low" at E.
3. Target a "major swing high" prior to the "crown" pattern formation.

Chapter 10: Price-Action

10.1. Cup and Handle Pattern

Cup and Handle Pattern

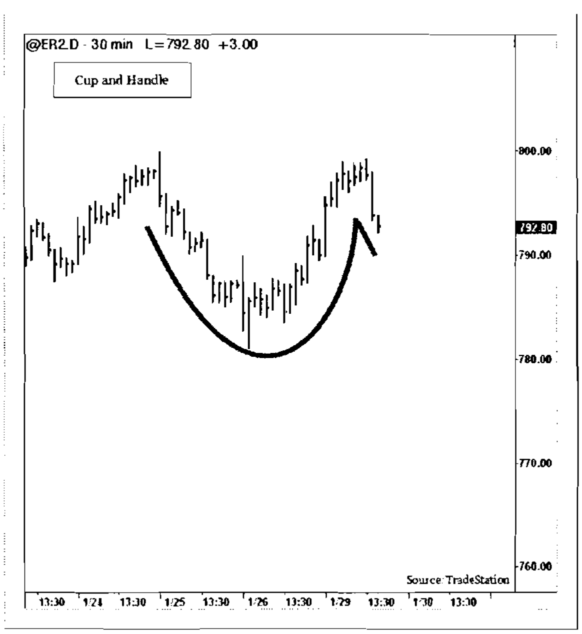
"Cup and Handle" is a continuation pattern that usually forms in bullish markets. Most "Cup and Handle" patterns are very reliable and offer great trading opportunities. "Cup" formation is developed during price rallies from the round bottom. The "Handle" part forms due to a price correction before a clear breakout to the upside.

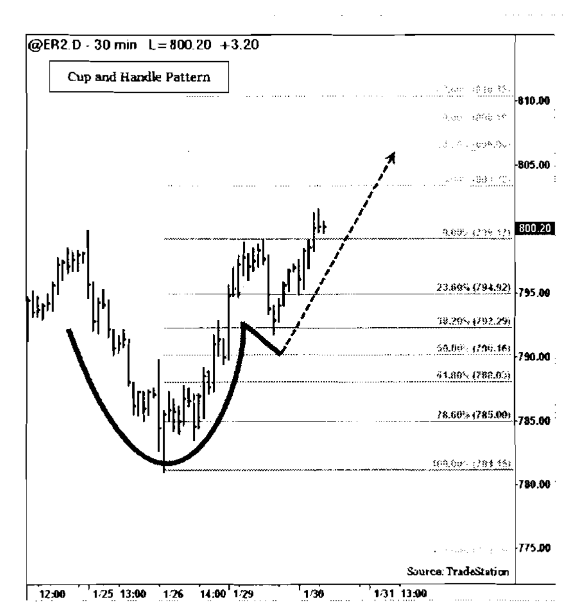
Trading "Cup and Handle" pattern is a two fold process. First a "Handle" can be traded as a corrective pattern and secondly as a continuation pattern (of the prior trend) after a breakout from the top of the "Cup." The handle part (right side) usually corrects about 25% to 38% (of depth of the Cup) from the high pivot point to the bottom of the Cup. After the correction (handle), the pattern will pick its prior trend and trade in a breakout fashion above the 'top' of the 'Cup'.

Trade: The best trading opportunity is generated when the price clears the top of the "Cup." A "long" trade is entered when the price closes above the high of the cup.

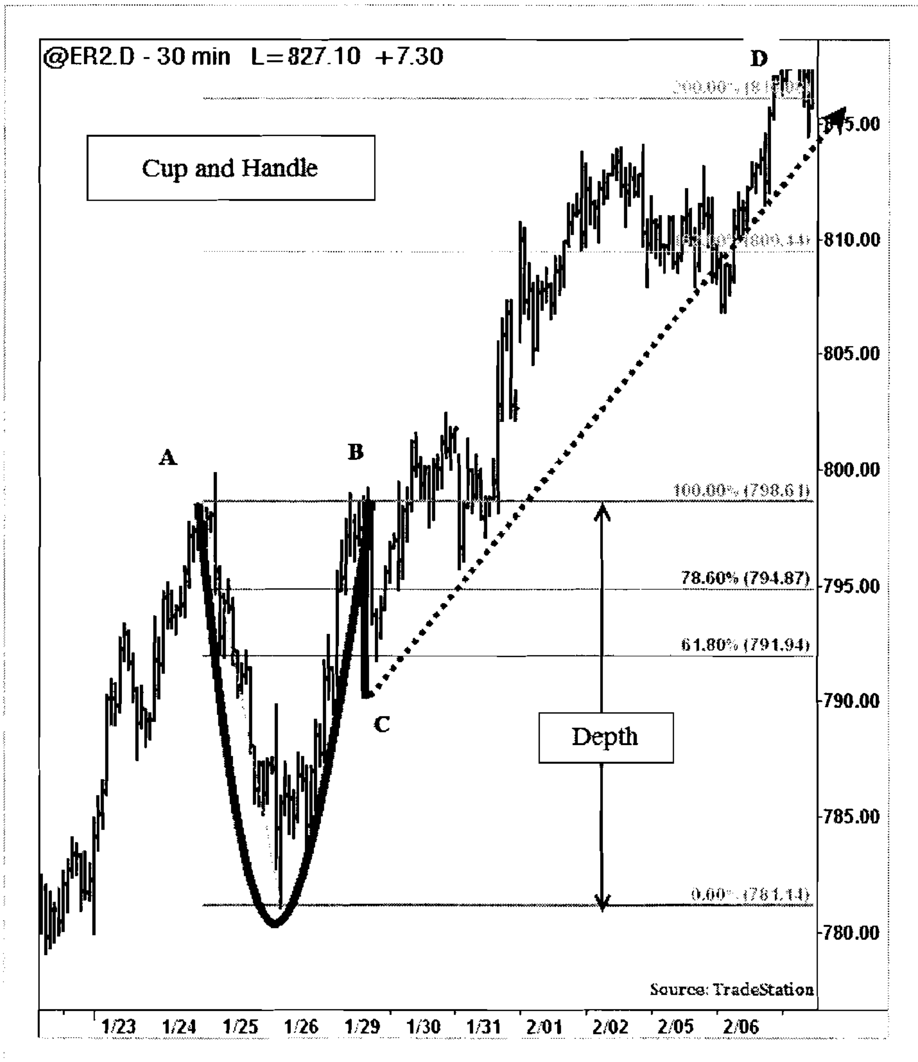
Target: Most "Cup & Handle" patterns result from 62% to 100% the depth of the "Cup" from the breakout levels.

Stop: Place a "stop" order below the low of the handle to protect the trade.





Trading Cup and Handle Pattern

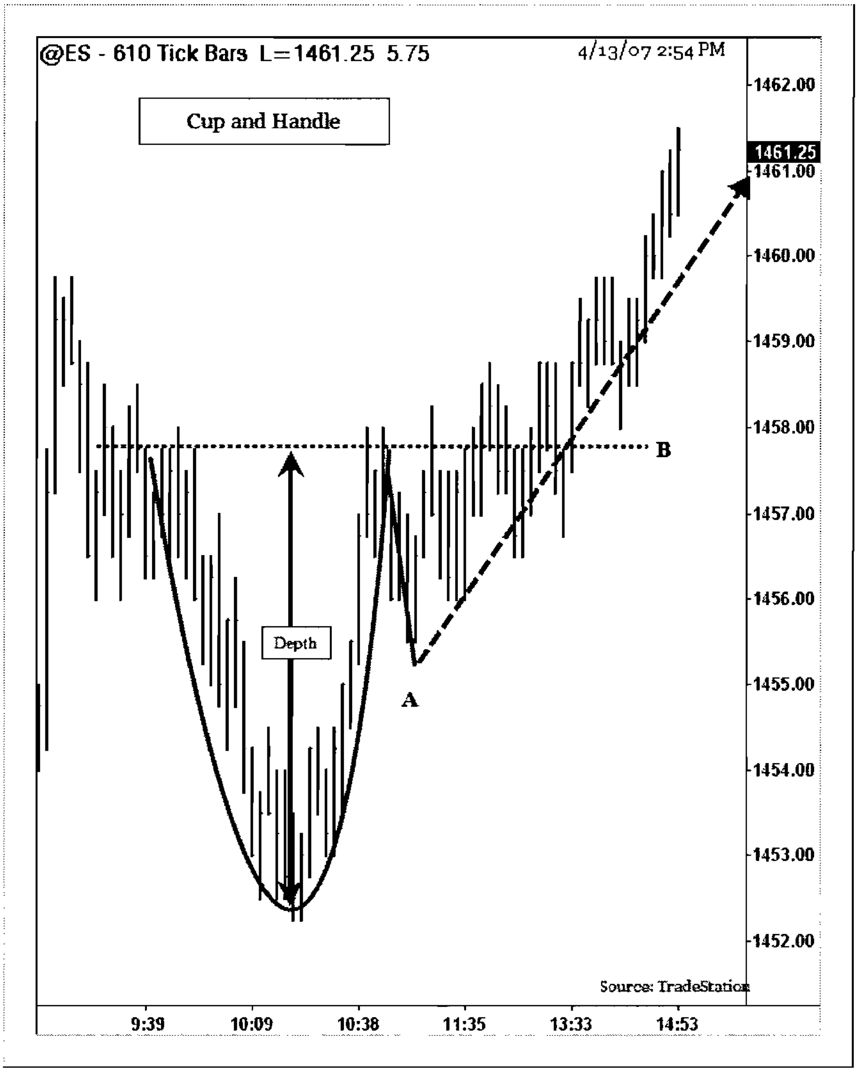


Trading Cup and Handle Pattern

The example above shows a "Cup and Handle" pattern formation from the Russell Emini (ER2) 30 minute chart. In January 2007, a "Cup and Handle" pattern has formed after ER2's uptrend. During the Cup formation (January 24 to January 29) the "Cup" pattern made highs at 798.6. A "Handle" part is formed from January 29 to January 30th between798 to 792. On January 31st, prices closed above the high of the "Cup" level at 799 to signal a "long" trade.

1. Enter a "long" trade above the top of the "Cup" level at 799.
2. Place a "stop" order below the "Handle" at 792.
3. Place a target at 62% to 100% of "cup" depth (12 to 18 points) from the breakout.

Trading Cup and Handle Pattern



Trading Cup and Handle Pattern

The example above illustrates a "Cup and Handle" pattern from the S&P Futures (ES) 610 tick chart. On April 13, 2007, during the morning session, ES formed a "Cup" formation between 1452 and 1453. Around 10.38 a.m., a correction in the trade resulted in a 38% "Handle" formation to the 1455.5 level. A breakout above the top of the "Cup" level signaled a "long" trade.

1. Enter a "long" trade above the high of the breakout bar at 1458.
2. Place a "stop" order below the low of the "handle" at 1455.
3. Target from 62% to 100% depth of the "cup" above the breakout level (1461 to 1464).

10.2. Head and Shoulders

Head and Shoulders Pattern

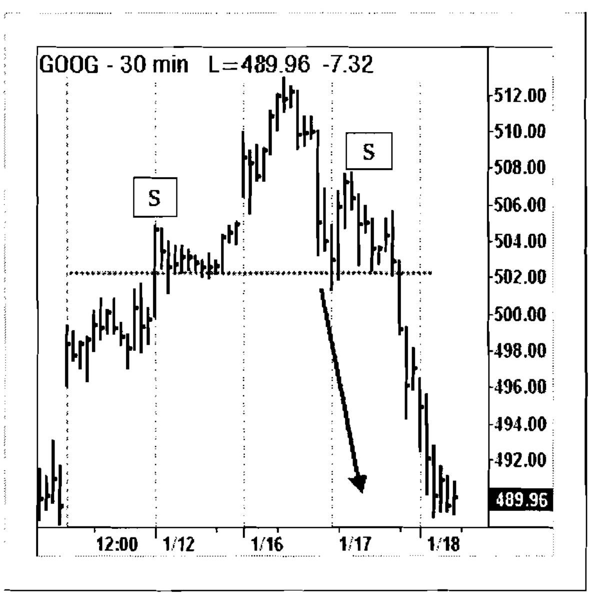
"Head and Shoulders" patterns are reversal formations that usually form at the market tops. "Head and Shoulders" patterns are very reliable, but failures do occur. When Head and Shoulders patterns fail, they reverse the pattern and trade in an explosive manner. Most "Head and Shoulders" patterns can be detected using volume patterns. During the left shoulder and the beginning of the "Head" formations, the volume will be heavy. While forming, volume dissipates on the right shoulder, and the volume increases during the breakdown.

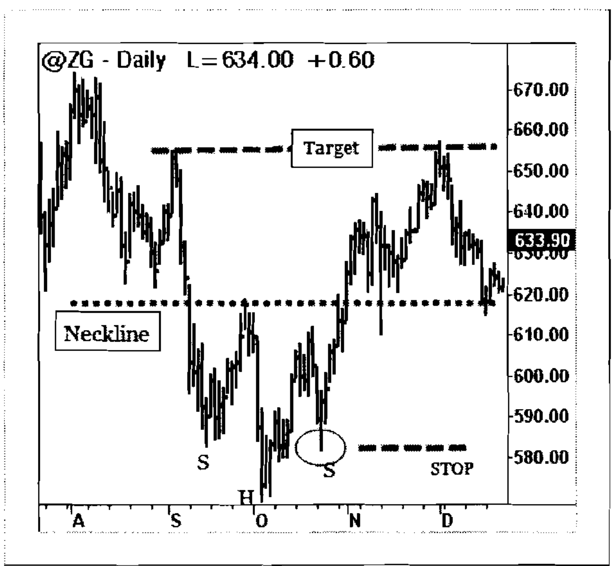
A trend line or neckline is drawn connecting the "Head and Shoulders" pattern to determine the potential trade opportunities and targets. The neckline can be also formed in an angle (slanted).

Trade:
Connect "Head and Shoulders" bottoms in a trend line or neckline. When the price closes below the neckline, a potential short trade is signaled. Short one tick below the breakdown bar's low.

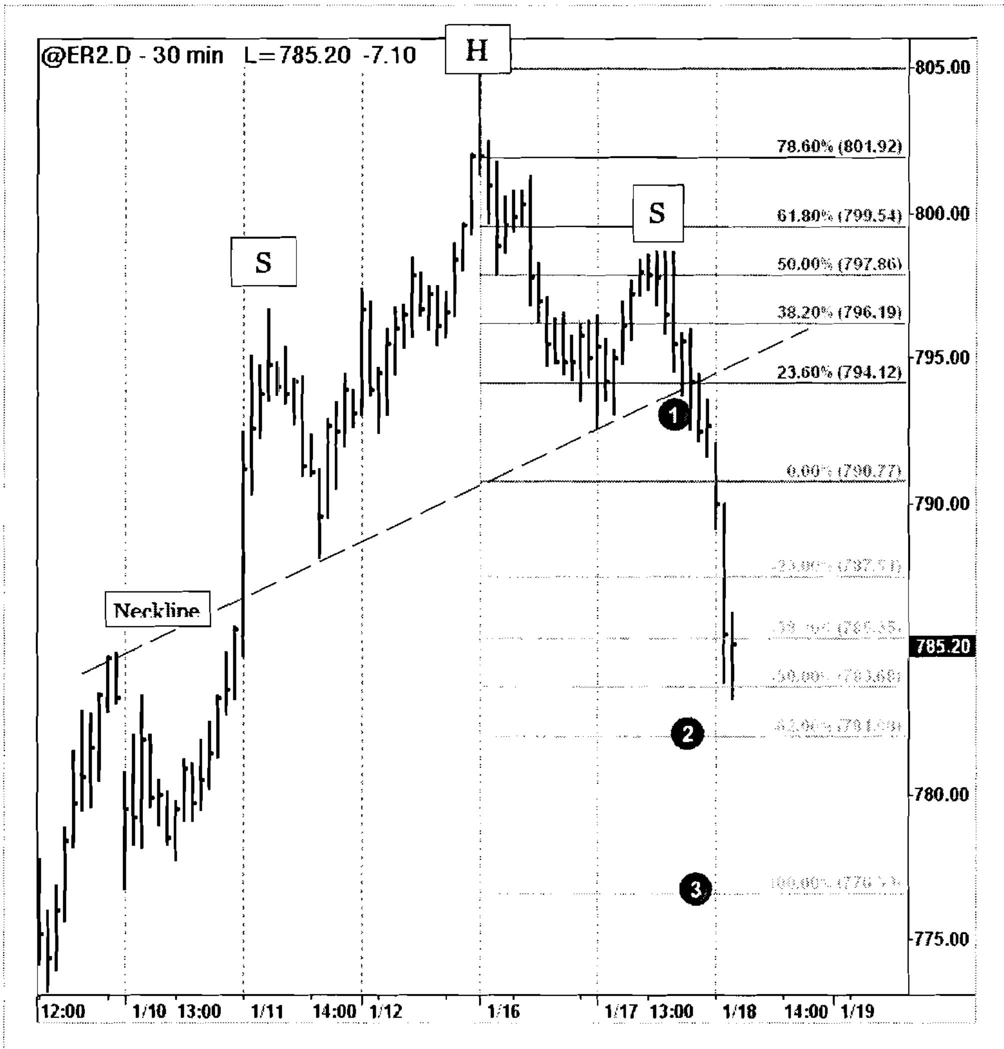
Target: Compute the vertical distance between the "apex" of the "Head and Shoulders" pattern and the neckline. The target is set below this distance from the neckline.

Stop:
After a trade entry, if the price closes above the neckline, a potential failure of the pattern is signaled. Place a "stop" order above the neckline.





Trading Head and Shoulders Pattern



Trading Head and Shoulders Pattern

The example above shows a "Head and Shoulders" pattern formation from the Russell Emini (ER2) 30 minute charts. From January 11 to January 17, 2007, the pattern formed "Head and Shoulders" pattern. A neckline was drawn connecting the bottom of the two "shoulders" to signify the potential breakdown trade. On January 18, ER2 closed below the neckline to confirm a breakdown.

1. Enter a "short" trade below the low of the breakdown bar at 794.
2. Target 62% to 100% of the pattern depth (14 points) from the breakdown level at 794.
3. Place a "stop" order above the neckline at 796.

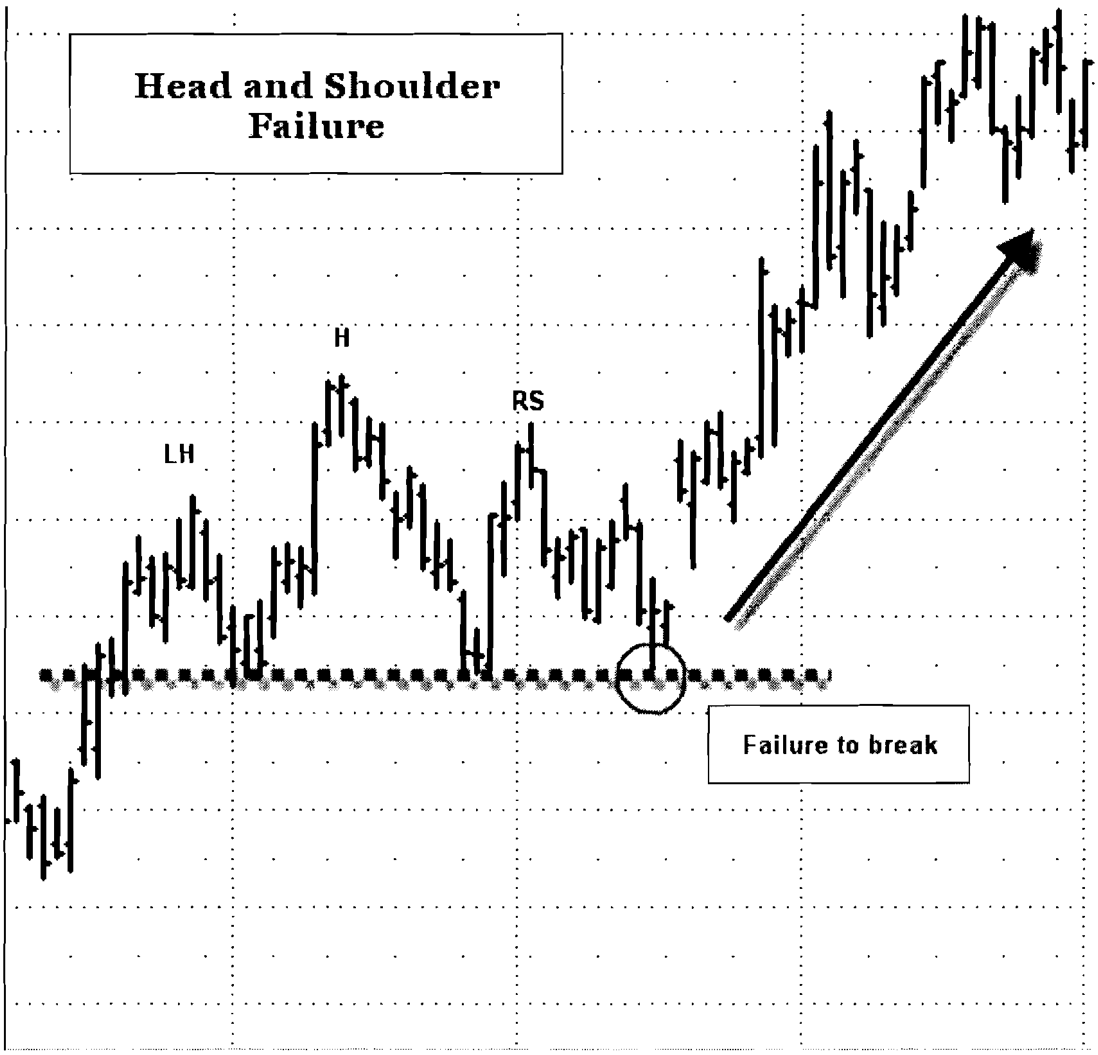
Trading Inverse Head and Shoulders

<u>Trading Inverse Head and Shoulders Pattern</u>

The example above illustrates the "Inverse Head and Shoulders" pattern formation from Microsoft Corporation's daily chart. In anticipation of Microsoft's Vista Operating System, investors drove the stock upside from August to December 2006. A neckline is drawn connecting the "Head and Shoulders" swing highs. In late July, Microsoft traded above the neckline and signaled a potential long trade.

1. Enter a "long" trade above high of the breakout bar at $24.
2. Place a "stop" below the right shoulder at $22.
3. Target the depth of the "Head and Shoulders" pattern ($3) from $24 to $27.

Trading Head and Shoulders Failure

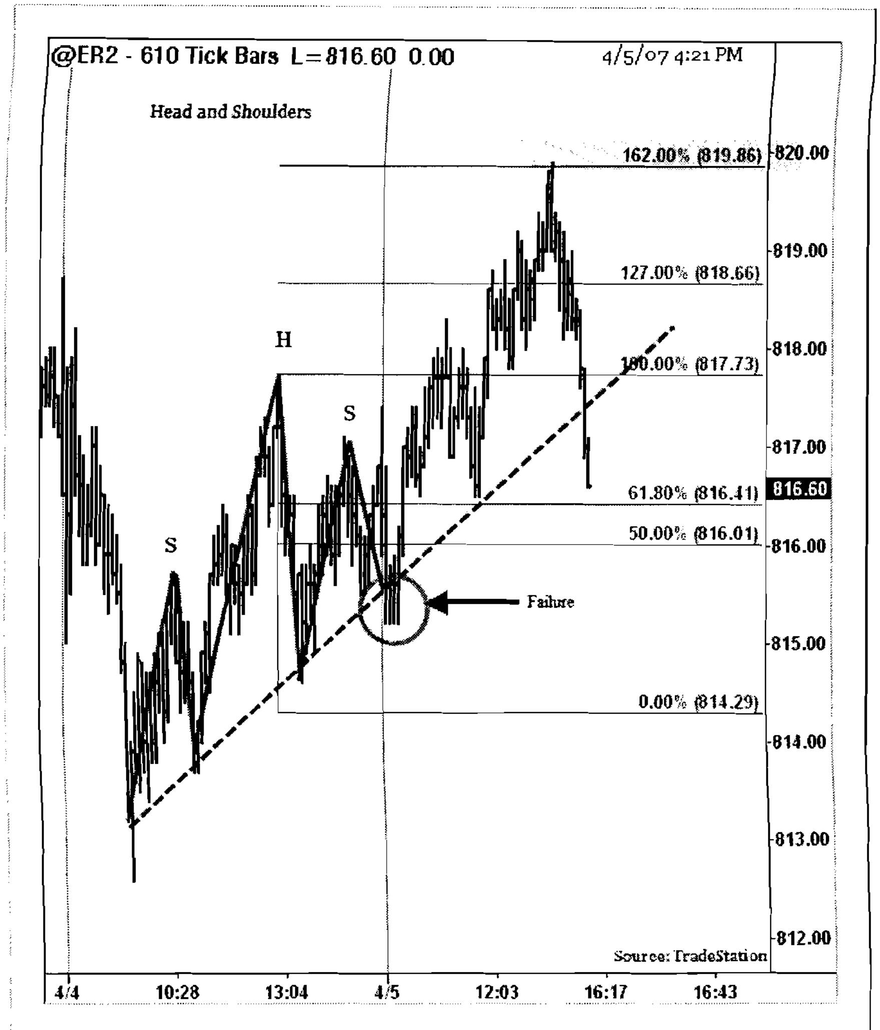


Trading Head and Shoulders Failures Pattern

The chart above illustrates an example of a "Head and Shoulders" pattern failure. A trend line is drawn connecting the bottoms of the "Head and Shoulders." The "Head and Shoulders" failures occur when prices failed to close below the trend line.

1. Enter a "long" trade above the "high" of the right side shoulder.
2. Place a "stop" order below the neckline for trade protection.
3. Target the depth range from head to the neckline from the trade entry.

Trading Head and Shoulders Failure



<u>Trading Head and Shoulders Failures Pattern</u>

The example above illustrates a "Head and Shoulders" pattern failure from the Russell 2000 Emini chart. On April 4 2007, ER2 formed a "Head and Shoulders" pattern with inclined (slanted) neckline. On April 5 2007, a trend line breakdown is anticipated for a "Head and Shoulders" pattern. This pattern is an example of reversal formation as "Head and Shoulders" pattern failed and continued in a reverse direction from the previous trend.

1. Enter a "long" trade above the high of the right side shoulder.
2. Place a "stop" below the low of the prior swing before the trade.
3. Target 100% of the depth of the "Head and Shoulders" pattern from the trade entry.

10.3. Parabolic Arc

Parabolic Arc Pattern

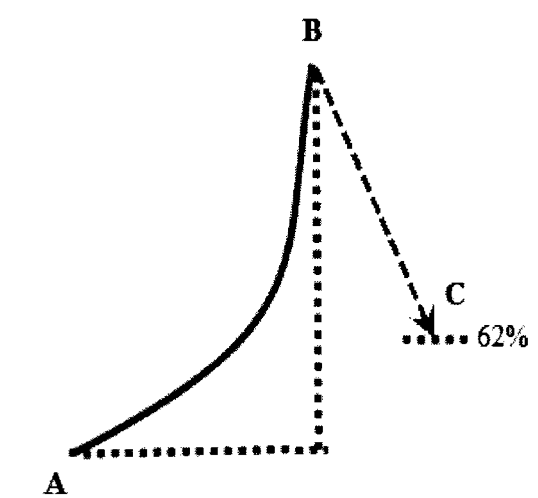
"Parabolic Arc" patterns are very rare, but occur in mega-bull trends. These patterns form in bull markets where irrational buying in the public generates a strong rally as the prices rise almost vertically, e.g., the Internet Boom in 2000 and the Metals boom in 2005-2006. Most of these "Parabolic Arc" patterns go straight up and then come straight down. Other examples of this market types are the NASDAQ bullish markets during 1999-2000 and Gold prices from 1977 to 1982.

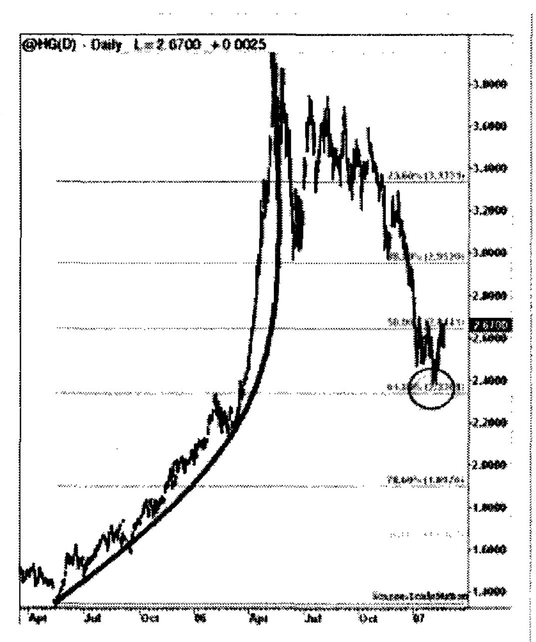
"Parabolic arc" is a reversal pattern and has a very predictable outcome. Although they are predictable, they are relatively difficult to trade since the market sentiment is bullish and may be relatively tough to point reversals to trade. Most "Parabolic arc" patterns have a significant correction of 62% from the top.

Trade: After a parabolic base, prices move up vertically and eventually the acceleration comes to a stop and then reverses. Prices start showing lower-lows and may attempt to regain the top again. Enter a "short" trade at the second failed attempt to test the peak or at the trend line breakdown connecting the major swing lows.

Target:
Measure the distance of the rise from the base to the top. Most "Parabolic Arcs" result in 62% retracement.

Stop: Protect the trade by placing few ticks above the high point of the Parabolic Arc.





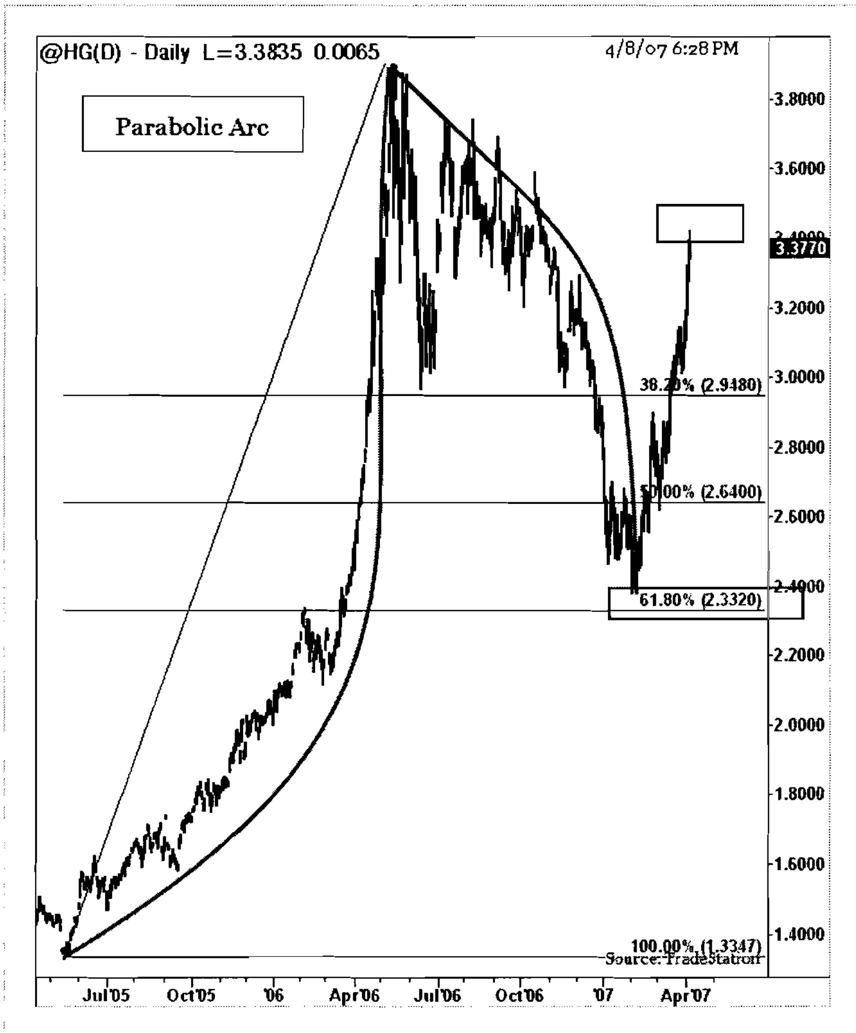
Trading Parabolic Arc

Trading Parabolic Arc

The example above illustrates a "Parabolic Arc" formation from the Silver futures chart. In 2005, Silver gained momentum as the metals sparked an interest in the world-wide markets. Silver futures rallied from the mid 7 to 15 range in a 12 month period. Prices rallied vertically during the beginning of 2006 setting up for a correction. From April 2006 to July 2006, silver futures corrected 62% to trade near 10.

1. Silver prices made a failed attempt to test previous "Swing High" (similar to 2B Sell setup).
2. Enter a "short" trade below first peaks' low at 14.50.
3. Place a "target" at 62% of the entire range to the first peak to 10.
4. Place a "stop" order above the close of the second peak at 15.5.

Trading Parabolic Arc



Trading Parabolic Arc Pattern

"Parabolic arc" trading is rare, but does occur in both extreme bull and bear markets. The chart above illustrates an example from the Copper futures daily chart. From the bottom of 1.4000, in July 2005 to April 2006, copper futures rallied to 3.8000. Copper futures retraced to 62% by January 2007 in a corrective "Parabolic arc" form to close near 2.4000. The "Parabolic Arc" resulted in another corrective phase (to the upside) to bounce back to 62% level near 3.3800. Trading the parabolic curve is not an easy proposition, but long term trades may be profitable. Parabolic arcs also form in intra-day trading, displaying much more violent moves and may be difficult to trade.

10.4. Three Hills and A Mountain Pattern

Three Hills and A Mountain

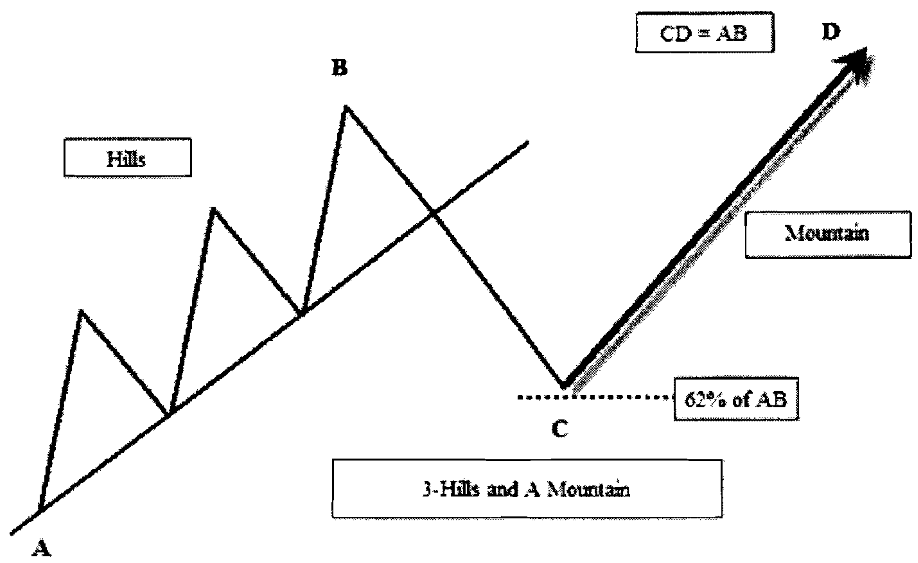
The "Three Hills and A Mountain" is rare, but a very reliable pattern. This pattern is similar to the "3-Drives (Bearish)" pattern. However, the "Three Hills" pattern is not of equal magnitude hills, and consists of another reversal trade in the original trend direction. The pattern also has a close relation to Elliott wave theory. The pattern supports Fibonacci retracement and extension concepts for key support and resistance areas. This pattern has two trade setups; one "short" and one "long."

Each hill retraces to a minimum of 0.5 to 0.618 (Fibonacci ratio) of its height before expanding into the next hill. After completion of the "Three Hills," a trend line is drawn in the chart below connecting the bottoms of the "Three Hills." A close below the trend line gives a short-trading opportunity. After completion of 62% of the prior range, it has another important reversal to the upside, the "Mountain" which rallies about 100% to 127% of entire hills range (AB).

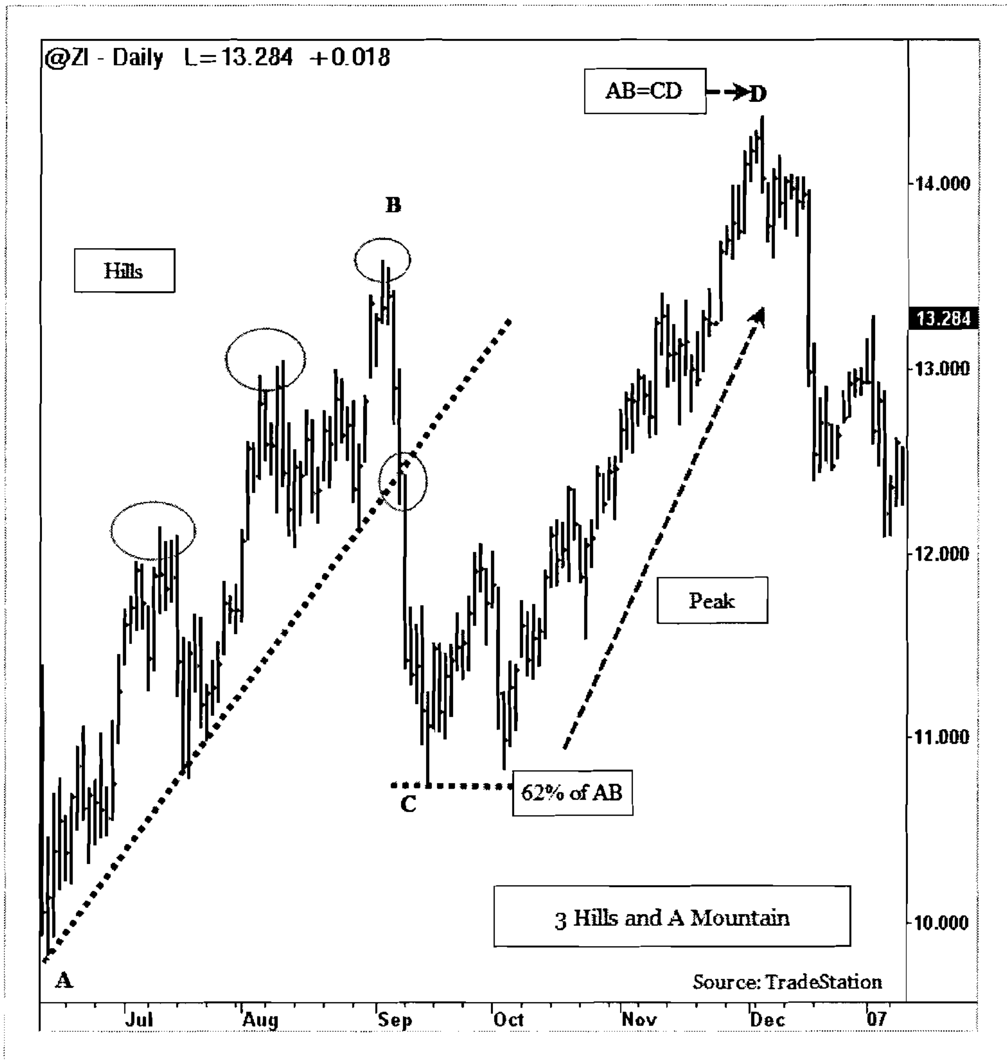
Trade: The first trade occurs on the close below the trend line. The second trade occurs after completion of a 62% retracement to the downside-"AB." The second trade is a "long" trade from "C."

Target: The first trade target (Short) is 62% of prior swing length-"AB." The second trade target is from the "C" level, which is 100% to 127% of "AB" to the upside.

Stop: Place the "first trade" stop above the "B" level and the "second trade" stop below the "C" level.



Trading Three Hills and A Mountain

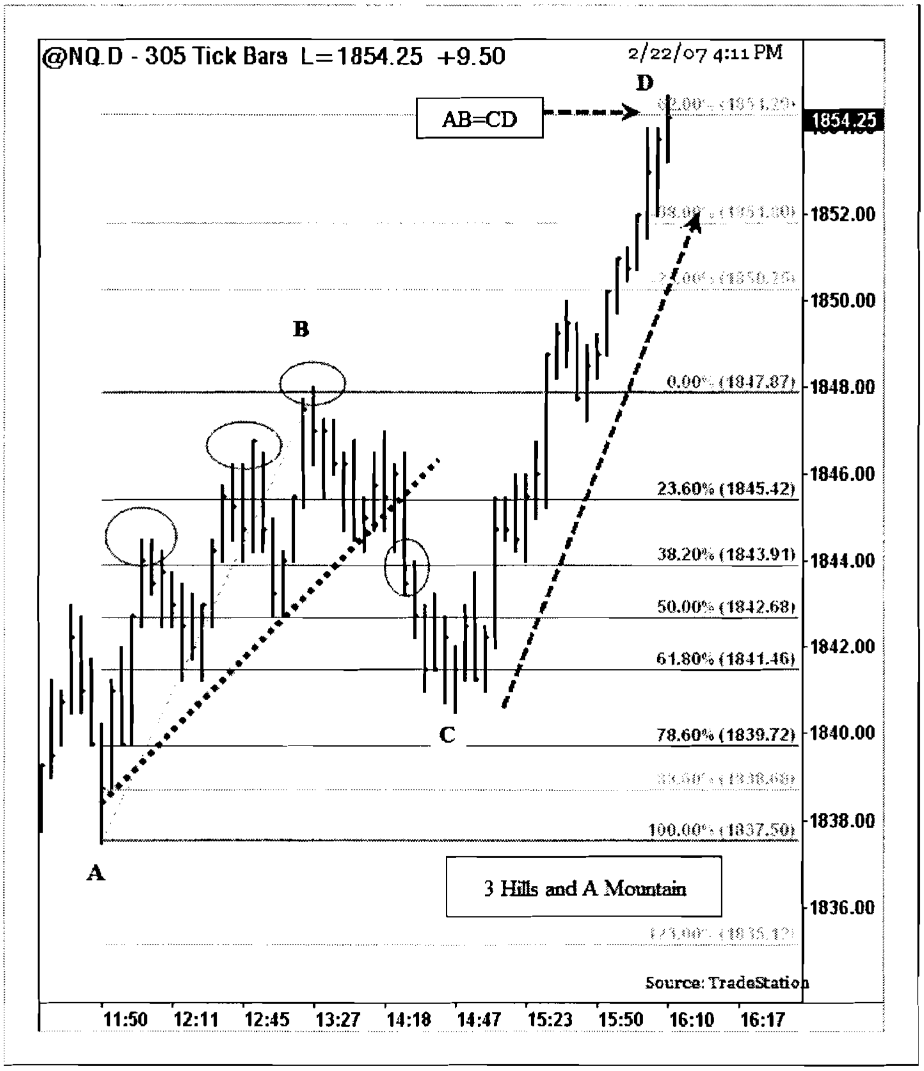


Trading Three Hills and A Mountain Pattern

The example above shows the "Three Hills and a Mountain" pattern from the daily Silver futures chart (ZI). From July 2006 to September 2006, Silver futures traded from 10.800 to 13.300 in a "Three Hills and A Mountain" pattern. A trend line was drawn in the chart above connecting the bottom of the "Three Hills." In September 2006, Silver futures closed below the trend line at 12.000, signaling a potential short-trade. The short trade target is set for 62% of "AB" at 10.975.

1. After reaching level "C"at 62%, a "long" trade is triggered above the previous bar's high at 11.500.
2. The target is set at AB=CD range at 14.400.
3. Place a "stop" order below level "C" at 10.970.

Trading Three Hills and A Mountain



Trading Three Hills and a Mountain Pattern

The chart above illustrates a "Three Hills and A Mountain" pattern from the NASDAQ futures (NQ) 305 tick chart. On February 22, 2007, around noon, the pattern started to emerge as NQ made "Three Hills" and with Fibonacci proportionate retracements. A trend line is drawn connecting the three hills for a potential breakdown. At around 2.15 pm, NQ futures closed below the trend line and retraced to 62% of AB at level "C."

1. After confirming level "C" at 62% of AB, a "long" trade is entered above previous bars' high at 1844.
2. The target is set at "AB" length from level "C" at 1854.
3. A "stop" order is placed below level "C" at 1840.

10.5. Three Valleys and A River Pattern

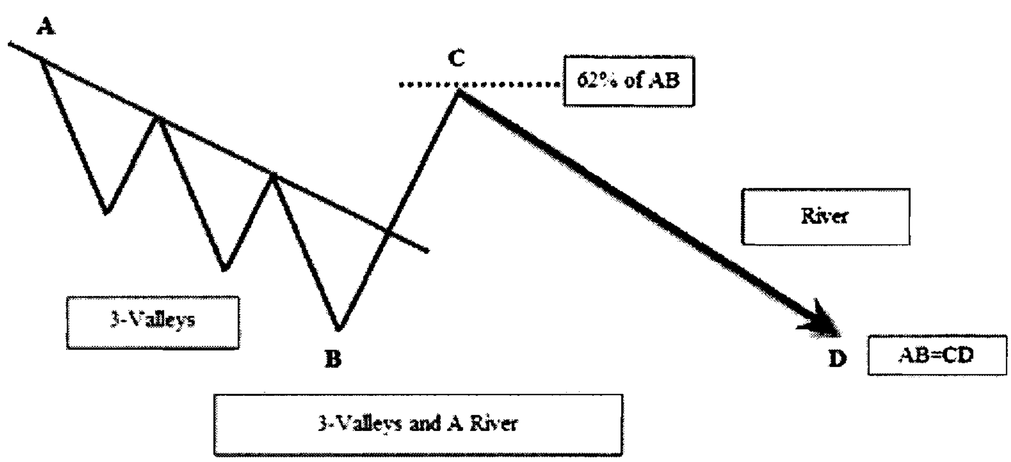
Three Valleys and A River

The "Three Valleys and A River" pattern is similar to the "Three Hills and A Mountain" pattern but in a reverse sense, and it is also similar to the "3-Drives (bullish)" pattern. The "Three Valleys and a River" pattern is very rare, but it is also very reliable pattern. It does have 3 distinct valleys as prices making three thrusts to go up and succeed in the third attempt. Usually this pattern occurs near the market bottoms. The "Three Valleys and a River" pattern further supports Fibonacci ratios among the three drives. Each valley retraces to a minimum 0.5 to 0.618 Fibonacci ratio of its height before expanding into the next valley. After completion of three valleys, a trend line is drawn connecting the top of the three valleys. Prices closing above the trend line signal a "long" trade. After completion of 62% of the previous range, it has another significant reversal to the downside (River).

Trade: The "Three Valleys and a River" pattern presents two trade setups. First trade: A close above the trend line signals a reversal. Enter a "long" trade one-tick above the high of the breakout bar. Second Trade: After completion of 62% of the AB range, enter a "short" trade at point "C."

Target: Target setups for the above two trades. The first trade setup is "long" from the three valleys. For this trade setup, set a target at 62% to 78% of the AB range. The second trade results in 100% of the AB range retracement from point "C."

Stop: For the first trade, if prices close below the trend line, close the trade. For the second trade, if prices close above level "C", close the trade.



Trading Three Valleys and A River

Trading Three Valleys and A River Pattern

The example above shows a "Three Valleys and a River" pattern from the ZLC weekly chart. ZLC established a 3-Valley pattern from July, 2005 to April, 2006. A trend line is drawn the connecting the tops of three valleys. In June 2006, ZLC closed above the trend line to confirm a "long" trade.

1. Enter a "long" trade above the high of the breakout bar from the trendline.
2. Place a "stop" order few ticks below the trend line.
3. Target 62% of the range between the top of the first valley to the bottom of the third valley.

Trading Three Valleys and A River

Trading Three Valleys and A River Pattern

The chart above illustrates an example of "Three Valleys and A River" pattern from Tyco's (TYC) weekly chart. From April 2005 to October 2006, TYC formed three "valleys" and attempted to rise three times beyond $30. A trend line is drawn in the chart above connecting the three valley tops for a trade setup. In October 2006, Tyco closed above the trend line to signal a potential long setup.

1. Enter a "long" trade above the high of the breakout bar at $27.
2. Place a "stop" order below the trend line to protect the trade at $25.5.
3. Place a "target" at the 62% range of the all three valleys.

10.6. Spike and Ledge Pattern

Spike and Ledge Pattern

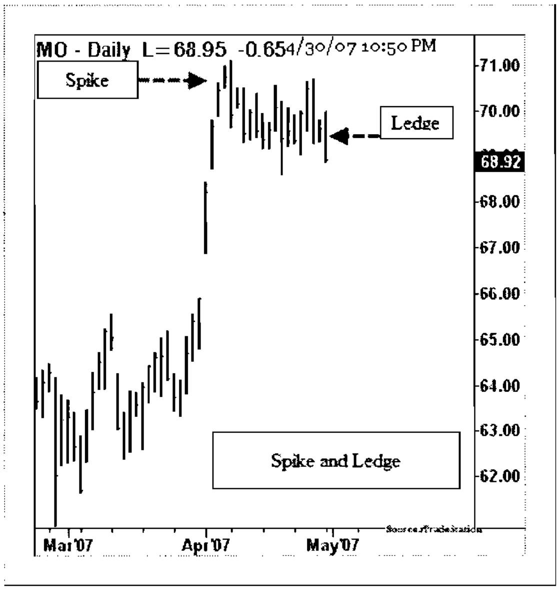
The "Spike and Ledge" pattern was first defined by Larry Connors and Linda Raschke in the book, *Street Smarts*. As the name suggests, markets make a "climax high" or "climax low" (New High or New Low) spikes followed by a resting point or "ledge" then a reversal of the prior trend.

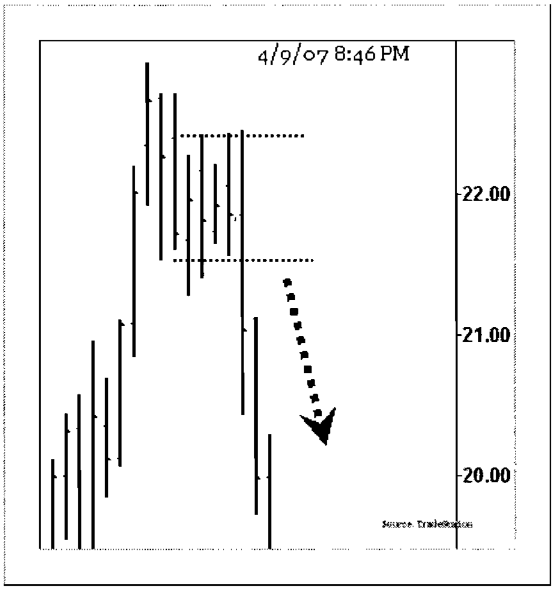
A "Ledge" pattern is defined as a series of bars with matching lows and highs within a certain number of bars, namely 20. Usually "ledge" patterns are considered as a resting point before markets pick a clear direction, which is mostly a reversal of the current major trend. Ledge patterns are effective in all time-frames but they are more effective during intra-day trading.

Trade: Set a trade entry at the "breakout" or "breakdown" from the "Ledge" formation. Wait for a clear breakdown or breakout and enter at the low (or high) of this bar.

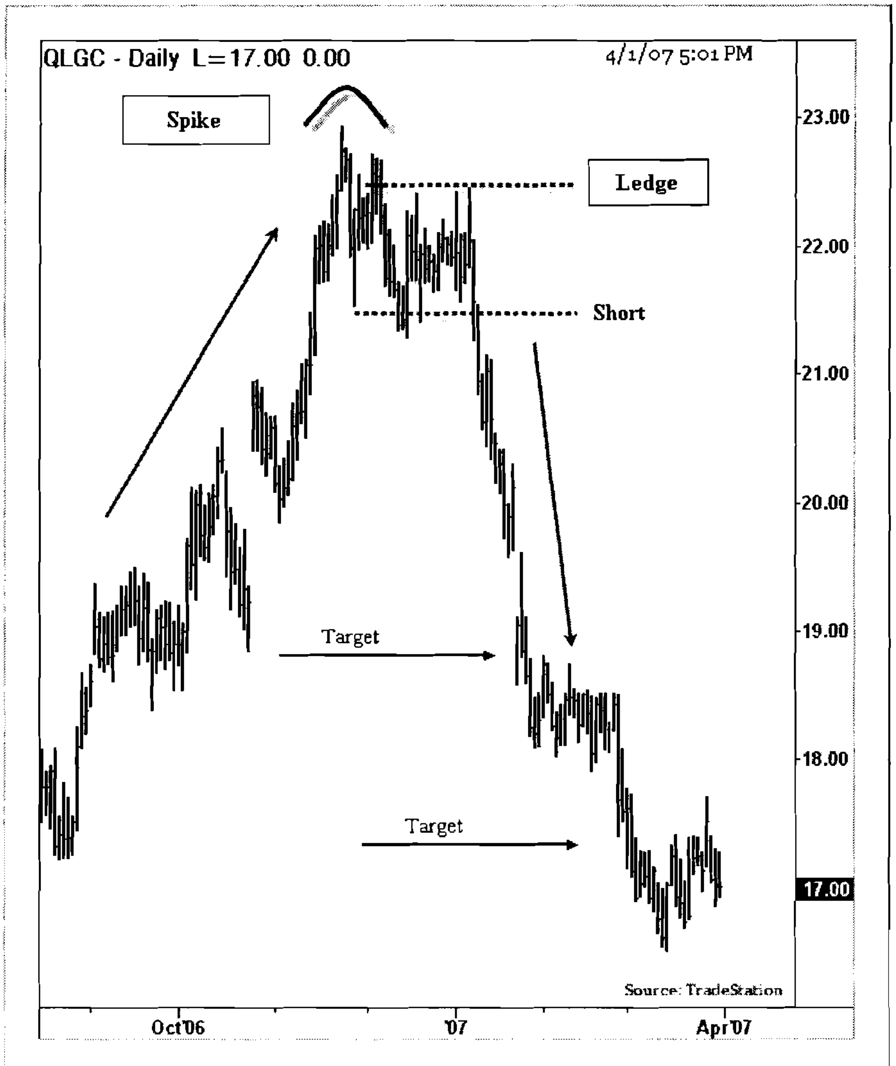
Stop: Place a "stop" order on the opposite side of the "Ledge" pattern. For long trades, place a "stop" order below the low of the "Ledge" pattern. For "short" trades, place a "stop" order above the high of the "Ledge" pattern.

Target:
Usual targets would be the prior "major swing high" for long trades or prior "major swing low" for short trades from the "breakdown" or "breakout" levels. If markets experience a "gap" prior to the "Spike" formation, targets after a breakdown/breakout from the "Ledge" pattern would be the "gap" levels. Otherwise, the target would be a prior "swing low" or prior "swing high" before the "Spike" formation.





Trading Spike and Ledge Pattern

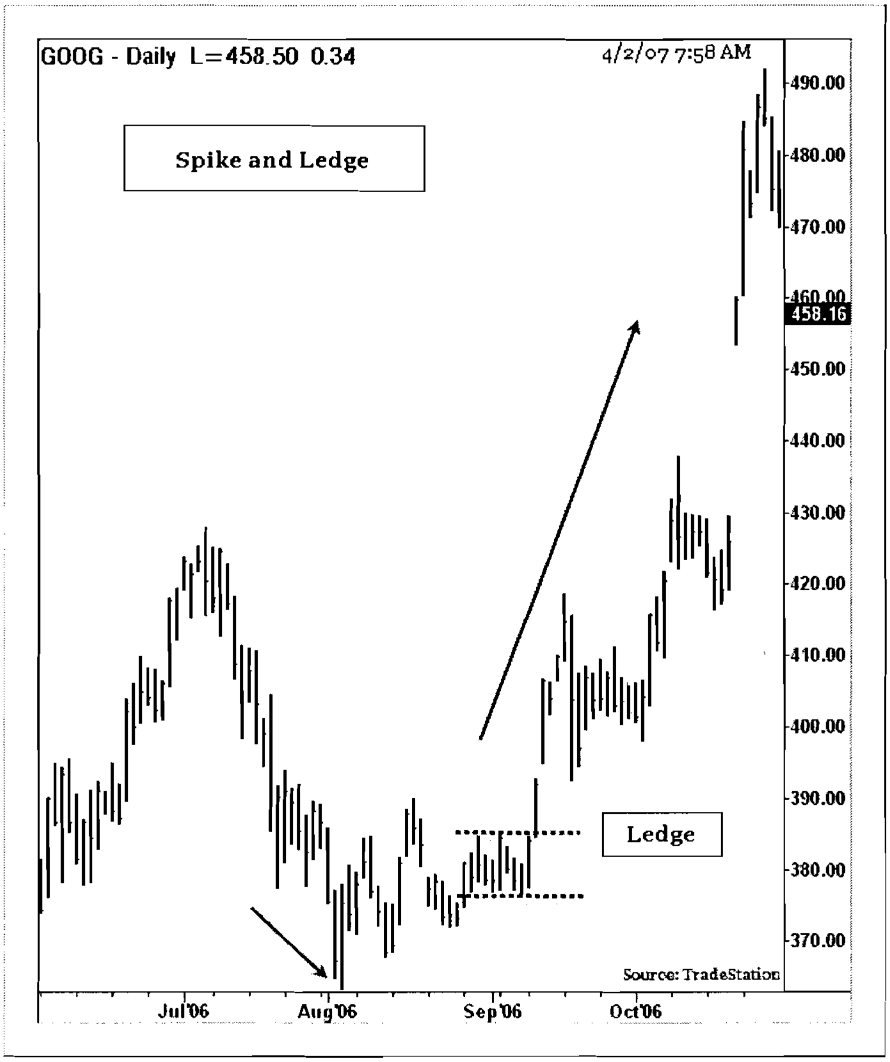


Trading Spike and Ledge Pattern

The example above illustrates a "Spike and Ledge" pattern from the QLogic (QLGC) daily chart. QLogic made a new "high" in November, 2006 and closed near $23. The last few days in November, 2006 were of high volume trading. After making a new high, QLogic attempted to sell-off and formed a "Ledge" pattern with matching highs and lows. In January 2007, QLogic broke out of the "Ledge" pattern and presented a "short" trading opportunity.

1. Enter a "short" trade below the low of the breakdown bar at $21.50
2. Place a "stop" order above the high of the "ledge" pattern at $22.50
3. Target the major "swing low" prior to the "spike" at $19.

Trading Spike and Ledge Pattern



<u>Trading Spike and Ledge Pattern</u>

The chart above demonstrates a "Spike and Ledge" pattern from the Google daily chart (GOOG). After a sell-off in July 2006, GOOG made a climax sell-off to close below 365 with high volume. In September 2006, GOOG attempted to rally back and made a "Ledge" formation with a series of matching highs and matching lows. A breakout above the "Ledge" formation signals a "long" trade to the upside.

1. Enter a "long" trade above the high of the breakout bar.
2. Place a "stop" loss below the "low" of the "Ledge" pattern.
3. Place a "target" at "swing high" prior to the down "Spike."